

W1 UNIT + MEDIA MAP
Digital media computing: acquire → digitise → process → understand → compress → deliver; exam favours principles + assumptions + trade-offs, not names only.
Media stack: audio: waveform/spectrum; image: pixels/intensity/colour; video: frames + time; graphics: geometry/material/camera; 3D/VR/AR: pose, depth, interaction.
Answer pattern: problem → cause → course method → parameters → limitation/evaluation.
Signal: continuous physical quantity; **sample:** value at discrete time/space; **quantise:** map amplitude to finite levels; **encode:** bit representation + container/metadate.
Pipeline design: identify user need; choose representation; pre-process; extract features/detect; post-process; compress/store; evaluate; state missing assumptions.
Common exam traps: omit units, confuse bits/bytes, assume fps/duration, treat enhancement as recognition, ignore latency/storage/privacy/bias.

Goal	Typical metric
readable board	OCR acc., contrast, blur score
detection	mAP, precision, recall, IoU
compression	bpp, PSNR/SSIM, MOS
real-time	fps, latency, memory

Open-book strategy: put formulas + decision rules; in answer, explain why method fits scenario.

W2 DIGITAL BASICS
Sampling: spatial/time discretisation; Nyquist: sampling freq. $f_s \geq 2f_{max}$ avoids aliasing after anti-alias filtering.
Aliasing: high freq masquerades as low freq when undersampled; fix by higher f_s , optical/low-pass filter, smaller motion per frame.
Quantisation: b bits = 2^b levels; step $\Delta = (x_{max} - x_{min}) / (L - 1)$; error $\epsilon \propto \Delta/2$, $\Delta/2$ if uniform.
Storage: bytes = width $\times$ height $\times$ frames $\times$ bytes/pixel; frames = fps $\times$ seconds. Audio bytes = $f_s \times b / 8 \times channels \times seconds$.
Bit depth: 8-bit grayscale = 256 levels; RGB24 = 3 bytes/pixel; alpha adds transparency; HDR uses more bits/dynamic range.
Dynamic range: ratio of max/min measurable intensity; clipping loses highlights/shadows; tone mapping compresses range for display.
Assumption check: if size question lacks fps, duration, active resolution, chroma format, bit depth, compression ratio, state missing info before calculation.

Format	Y	Cb/Cr	bytes/pixel
RGB 4:4:4	1	3	@8b/ch
YCbCr 4:4:4	1	1+1	3
4:2:2	2/2px	1+1/2px	2
4:2:0	4/4px	1+1/4px	1.5

Psycho-visual: HVS more sensitive to luminance detail than chroma; basis for YCbCr, chroma subsampling, perceptual quantisation.

W3 IMAGE PROCESSING I
Pixel ops: negative $s = L - 1 - r$; $\log s = c \log(1 + r)$ expands dark; gamma $s = c r^{\gamma}$; threshold $g = 1 \{f \geq T\}$.
Histogram: $h(r_k) = n_k$; pdf $p(r_k) = n_k / N$; CDF $C(r_k) = \sum_{j \leq k} p(r_j)$.
Hist. equalisation: $s_k = \text{round}((L - 1)C(r_k))$; map each old level r_k to s_k ; improves global contrast but may amplify noise.
HE calculation: count levels → pdf → cdf → transform table → replace pixels; keep repeated levels identical.
Convolution/correlation: output $g(i, j) = \sum_m \sum_n w(m, n)f(i + m, j + n)$ if mask applied as written; true convolution flips kernel.
Filters: mean/Gaussian smooth noise but blur edges; median removes impulse noise; sharpening/high-pass boosts detail/noise.
Sobel $G_x = \begin{bmatrix} -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$, $G_y = \begin{bmatrix} 0 & 0 & 1 \\ -1 & -2 & -1 \end{bmatrix}$, $|\nabla I| \approx |G_x| + |G_y|$.

Interpretation: large $|G_x|$ = strong vertical edge; sign tells dark-to-bright direction under chosen mask; G_y for horizontal edge.

W4 IMAGE PROCESSING II
Enhancement vs restoration: enhancement improves visual/useful quality; restoration models degradation/noise.
Noise: Gaussian: sensor/read noise; salt-pepper: impulse; Poisson: photon count; motion blur: exposure/camera/object movement.
Preprocess choice: noisy edge detection → Gaussian/median then Sobel/Canny; low contrast → HE/LAHE; colour cast → white balance/colour constancy; perspective → homography.
Geometry: affine preserves parallel lines; projective/homography maps planes; interpolation: nearest/fast/blocky, bilinear smoother, bicubic higher quality.

Homography: $\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$, use 4 point pairs for planar board rectification.
Morphology: erosion shrinks bright fg/removes specks; dilation expands/bridges; opening = erosion+dilation; closing = dilation+erosion.
Segmentation: global threshold simple; Otsu maximises between-class variance; adaptive threshold handles uneven illumination; region growing needs seed/similarity.
Boundary cases: threshold fails w/ shadows; smoothing hurts thin text; HE on already high-contrast image may over-enhance;

interpolation changes pixels.

Task	Method	Risk
document photo	homography+CLAHE	wrong corners
edge map	denoise+Sobel/Canny	blur detail
binary text	adaptive thresh.	broken strokes
noise removal	median	round corners

W5 IMAGE UNDERSTANDING I
Image understanding: infer objects/regions/actions/semantics from pixels; needs features/model + labelled/evaluated data.
Classical features: edges, corners, SIFT/HOG, Haar-like features; robust to some variation but hand-designed and limited for complex scenes.
Viola-Jones: Haar features + integral image + AdaBoost + cascade; fast frontal-face detector, weak for large pose/illumination/object variation.
Cascade: cheap early reject → progressively stronger classifiers; high speed but sensitive to distribution shift.
Segmentation vs detection: segmentation labels pixels/regions; detection outputs boxes/classes; classification labels whole image; recognition may include identity/action.
Detection metrics: precision = $TP / (TP + FP)$, recall = $TP / (TP + FN)$, IoU = $|B_p \cap B_g| / |B_p \cup B_g|$, mAP summarises PR over classes/thresholds.
Dataset plan: train/val/test split, stratify classes/scenes, avoid leakage from same video/frame, annotate boxes consistently, augment illumination/scale/blur/crop.

Confusion matrix: shows which classes are confused; precision matters for false alarms, recall for missed objects.

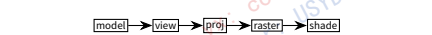
W6 IMAGE UNDERSTANDING II
CNN: learned convolutional filters + nonlinearities + pooling; hierarchy: edges → textures/parts → objects.
R-CNN: region proposals/selective search → warp each RoI → CNN per RoI → classifier + bbox regression; accurate but slow due repeated CNN.
Fast R-CNN: run CNN once on whole image → shared feature map → RoI pooling/align → cls + bbox heads; faster training/inference.
Detector families: two-stage (Faster R-CNN) offer accurate/localisation strong; one-stage (YOLO/SSD) faster end-to-end dense prediction; transformer detectors need data/compute.
Never say "always better" : choose by latency, hardware, data size, accuracy, occlusion, object scale, interpretability, update cost, failure impact.
BBox regression: predict offsets (t_x, t_y, t_w, t_h) refining proposals; NMS removes duplicate boxes using IoU threshold.
Pitfalls: mAP hides rare-class failure; high confidence can be wrong under domain shift; augmentation must match plausible deployment variations.



W7 VIDEO PROCESSING
Video: ordered frames; raw storage = frame bytes $\times$ fps $\times$ duration; temporal relation enables motion and compression.
Frame/shot/scene: frame = single image; shot = continuous camera take; scene = semantic group of shots; keyframe summarises shot.
Shot boundary: histogram diff, edge change ratio, optical flow discontinuity; hard cut abrupt, gradual transition needs temporal window.
Motion estimation: find displacement vector between blocks/regions; block matching minimises SAD/MAD/MSE. SAD $(u, v) = \sum_{x,y} |I_t(x, y) - I_{t-1}(x - u, y - v)|$; motion compensation predicts current frame from reference + residual.
Redundancy: temporal: adjacent frames similar; spatial: neighbouring pixels correlated; coding: symbols uneven; perceptual: HVS less sensitive to some errors.
Use carefully: motion blur/occlusion/illumination change break simple matching; global camera motion needs stabilisation/estimation; fast motion needs larger search.

Task	Idea	Exam phrase
video size	fps*time	state missing fps/time
compression	motion/residual	temporal redundancy
summary	shot/keyframe	avoid duplicate frames
tracking	flow/Kalman	handle occlusion

W8 GRAPHICS + ANIMATION
Graphics pipeline: model coords → world → camera/view → projection → rasterisation → shading/display.
Representation: raster: pixels good for images; vector: paths/scalable; mesh: vertices/edges/faces; point cloud: samples of 3D surface.
Transform: homogeneous coords enable translation/scale/rotation; compose matrices in correct order; perspective divide maps 3D to screen.
Animation: keyframes + interpolation; kinematics controls articulated motion; physics sim adds forces/collisions; motion capture data-driven.
Rendering: geometry processing → visibility/z-buffer → lighting/shading → texture mapping → anti-aliasing.
Lighting: ambient + diffuse (Lambert) + specular; normals affect appearance; texture maps add detail w/o geometry.
Trade-offs: realism vs fps; mesh detail vs memory; ray tracing quality vs compute; simplified shaders for mobile/AR.



W9 VISUAL COMPUTING ADV I
Content-aware resizing: preserve important content while changing aspect/size; seam carving removes/inserts low-energy connected seams.
Seam carving: compute energy (e.g. gradient magnitude) → dynamic programming cumulative cost → trace min vertical/horizontal seam → remove/insert → repeat.
DP vertical seam: $M(i, j) = E(i, j) + \min(M(i - 1, j - 1), M(i - 1, j), M(i - 1, j + 1))$.
Good for: backgrounds/textures; **bad for:** faces, straight structures, dense salient objects; protect/remove masks modify energy.
OCR/document capture: rectify board, denoise, normalise illumination, threshold/segment text, recognise, post-correct language; evaluation: character/word error rate.
HDR/fusion: combine exposures to recover detail; align images, estimate weights, fuse/tone-map; ghosting if motion.
Super-resolution: interpolation baseline; reconstruction multi-frame; DL learns mapping LR→HR but can hallucinate detail.

Problem	Classical	DL/hybrid
blur	deconvolution	deblur prior
low light	denoise+CLAHE	CNN deblur net
text region	MSER/threshold	detector+OCR
SR	bicubic	ESRGAN-like

W10 VISUAL COMPUTING ADV II
System design: exam scenario usually asks for robust pipeline under constraints; marks come from justified choices and limitations.
Board-capture pipeline: RGB→YCbCr/Lab; denoise; white balance; detect board corners; homography rectify; CLAHE on luminance; sharpen/threshold; detect text/diagram/equation; compress.
Evaluation pair: quantitative metric + qualitative user inspection; e.g. OCR CER + readability study, mAP + failure-case gallery, PSNR/SSIM + visual artefact notes.
Hybrid approach: classical pre/post-processing + deep detector/recognition; often best when data limited but robustness needed.
AR/VR/new media: registration, tracking, latency, calibration, depth/pose; user experience depends on stable frame rate and low motion-to-photon delay.
Ethics/deployment: privacy for captured media, consent, bias in training data, compression artefacts changing evidence, energy/compute on mobile.
Answer template: "I choose X because constraint A/B/C. Steps: ... Failure cases: ... Evaluation: ..."
Robustness knobs: augmentation (illumination, blur, scale, occlusion), confidence thresholds, fallback manual crop, domain-specific validation.

W11 MEDIA COMPRESSION
Compression: lossless preserves exact data; lossy discards less important info; rate-distortion controls bits vs quality.
Redundancies: spatial, temporal, spectral/transform, statistical/coding, perceptual.
JPEG: RGB→YCbCr; optional chroma subsample; split 8 $\times$ 8; DCT; quantise; zigzag; run-length + entropy code.
DCT compacts energy into low freq; quantisation step size controls loss. High-frequency coefficients often more coarsely quantised.
JPEG2000: wavelet transform, scalable quality/resolution, better at low bit-rate, supports ROI; more complex than baseline JPEG.
Video codec: intra prediction + inter prediction/motion estimation + transform residual + quantisation + entropy coding + loop filters.
Exam compare: JPEG block artefacts; JPEG2000 ringing/complexity; inter-frame compression efficient but errors may propagate.

Term	Meaning	Example
bpp	bits/pixel	rate
PSNR	log MSE quality	distortion
SSIM	structural similarity	perceptual-ish
CR	raw/compressed	storage gain

W12 PROJECT + SYNTHESIS
Project thinking: prototype design + media pipeline + feasibility + evaluation; final often integrates multiple topics in one application.
High-mark answer = concrete scenario-aware pipeline, not a list of algorithms. Tie every method to a problem and name a trade-off.
Camera app innovation: identify limitation (low light, blur, distortion, colour, storage); propose pipeline; explain compute/storage/IU; evaluate with real samples.
Model selection: classical if interpretable/small data/low compute; DL if complex variation + labels + hardware; hybrid when pre-processing improves data and model handles semantics.
Calculation mini-example: $720 \times 576 \times 4 \times 2$ @8b: 2 bytes/pixel = 829,440 B/frame; @25fps 40s: 829,440,000 B $\approx$ 791 MiB. RGB24 ratio = $3/2 = 1.5$.
Sobel mini-example: neighbourhood columns 10, 10, 80 repeated: $G_x = (-10 + 80) + (-20 + 160) + (-10 + 80) = 280$, $G_y = 0$, edge between cols 3/4.
HE mini-example: levels 0.3 counts (1, 1, 2, 4), $N = 8$, $L = 4$: CDF (0.125, 0.25, 0.5, 1) maps to (0, 1, 2, 3) by round $(3 \cdot CDF)$.

If asked...	Do this
compare methods	define each + 2 advantages + 2 limits
calculate	write formula; units, assumptions
design pipeline	steps + reason + failure + metric
critique claim	state false/conditional + constraints

Source scope: 2026 USYD outline W1-W12 + COMPS405 22/23/24/25 final/recall/review clusters.

FORMULA BANK

Unit conversions: 1 byte = 8 bits; 1 KiB = 2^{10} B, 1 MiB = 2^{20} B, 1 MB = 10^6 B if decimal requested; always state which.
Raw image: grayscale bytes = $W \times H \times b/8$; RGB bytes = $W \times H \times (3b)/8$; with alpha = $W \times H \times (4b)/8$.
Raw video: $WH \times \text{bytes/pixel} \times \text{fps} \times t$; data rate = frame bytes $\times \text{fps}$.
4:2:2 derivation: for 2 pixels store Y_1, Y_2, C_b, C_r ; @8b = 4 B/2 px = 2 B/px.
4:2:0 derivation: for 2×2 block store $4Y + 1Cb + 1Cr$; @8b = 6 B/4 px = 1.5 B/px.
Audio: PCM bytes = $f_s \cdot t \cdot channels \cdot b/8$; CD audio $44100 \times 16/8 \times 2 \approx 176.4$ kB/s.
RGB→CMYK: normalise R', G', B' ; $K = 1 - \max(R', G', B')$; $C = (1 - R' - K) / (1 - K)$ etc.; if $K = 1$, set $C = M = Y = 0$.
RGB→YUV intuition: Y: average brightness; U, V: colour difference; exact matrix depends on standard, so quote if supplied.
Thresholding: binary $B(x, y) = 1$ if $I(x, y) \geq T$ else 0; invert only if foreground convention says so.
Otsu: choose T maximising $\sigma_B^2(T) = \omega_0 \omega_1 (\mu_0 - \mu_1)^2$; assumes roughly bimodal histogram.
MSE/PSNR: $MSE = \frac{1}{N} \sum (I - K)^2$; PSNR = $10 \log_{10} (M \cdot A \cdot X^2 / MSE)$; higher PSNR usually less pixel error, not always perceptual.
IoU/NMS: keep highest-score box, suppress boxes w/ IoU above threshold; too low threshold loses close objects, too high keeps duplicates.

Symptom	Likely cause	First tool
low contrast	narrow histogram	HE/LAHE
grain/noise	sensor/high ISO	Gaussian/median/bilateral
blur	motion/defocus	deblur, faster exposure
colour cast	view angle	corner detect+homography
hugged video	lighting/WC	white balance/Lab/YCbCr
missed object	raw frames	subsample+codec
	model/domain shift	data+threshold tuning

Filter selector: mean smooths but blurs; Gaussian smooths weighted by distance; median preserves edges under salt-pepper; bilateral preserves edges but slower; Sobel detects gradients; Laplacian detects second derivative/edges but noise-sensitive.
Segmentation selector: threshold for bimodal intensity; adaptive for uneven illumination; edge/contour for object boundaries; watershed for touching objects; semantic segmentation for class-level pixel labels.
Detector selector: VJ if rigid/frontal/simple and CPU fast; HOG+SVM if moderate variation; Faster/Fast R-CNN if accuracy/localisation; YOLO/SSD if real-time; transformer if data/compute allow.
Compression selector: PNG/lossless for text/diagrams; JPEG for photos; JPEG2000 if scalability/ROI; H.26x/AV1-like for video; choose bitrate by acceptable artefact.
Critique template: define claim; identify hidden assumptions; compare alternatives; give scenario constraints; conclude conditional recommendation.

PAST-PAPER HOTSPOTS
Digitisation cluster: sampling vs quantisation; aliasing prevention; raw storage; audio/video size; missing assumptions.
Colour cluster: RGB/CMYK/YUV/YCbCr; luminance vs chroma; 4:4:4/4:2:2/4:2:0 storage quality trade-off.
Image processing cluster: HE calculation; threshold binary image; Sobel convolution; blur/noise solutions; enhancement vs edge detection.
Understanding cluster: VJ vs CNN detectors; R-CNN vs Fast R-CNN; training/evaluation plan; augmentation; precision/recall/mAP.
Video/compression cluster: motion estimation, shot/keyframe, temporal redundancy, JPEG/JPEG2000, codec redundancies.
Design cluster: camera app, board capture, OCR, HDR, seam carving, project-style integrated answer; reward = justification and trade-offs. For 25-style 4-question finals, expect: Q1 storage/colour/compression; Q2 detection selection; Q3 Sobel/HE/calculation; Q4 integrated media-system design.

WORKED MICRO-EXAMPLES
Video 4:2:0: 1920 $\times$ 1080 frame @8b 4:2:0 stores $W \times H \times 1.5 = 3, 110, 400$ B/frame $\approx$ 2.97 MiB.
RGB24 1 min: $1280 \times 720 \times 30 \times \text{fps}$ 60s RGB24: $1280 \times 720 \times 3 \times 1800 = 4, 976, 400$ B $\approx$ 4.64 GiB.
Audio: 3 min stereo 44.1kHz 16b: $44100 \cdot 180 \cdot 2 \cdot 2 = 31, 752, 000$ B $\approx$ 30.3 MiB.
Threshold: matrix values below 128 $\rightarrow$ 0, at/above 128 $\rightarrow$ 1; state if equality included.
Sobel constant patch: kernel positive and negative weights sum to 0, so constant region gives $G_x = G_y = 0$.
HE with absent levels: include zero counts in CDF; mapping remains monotonic; multiple old levels may map same new level.
R-CNN inefficiency: if 2000 proposals, original R-CNN does CNN 2000 times; Fast R-CNN shares conv once then processes RoIs.
mAP reasoning: high precision low recall = conservative detector; high recall low precision = noisy detector; tune threshold by application cost.
DESIGN ANSWER RUBRIC
Step 1 diagnose: list 3 concrete media problems (e.g. low light, perspective, blur) with technical cause and user impact.

Step 2 pipeline: representation conversion, denoise/enhance, geometry, segmentation/detection, compression, UI/fallback.
Step 3 justify: why each step fits one parameter or variant; one failure case; one mitigation.
Step 4 evaluate: quantify USYD各科往届题库 review; include edge cases and a table.
Example board capture: chroma homography fixes skew; adapt OCR/detector finds text/equation accuracy vs storage.
Example conveyor sorting: de-robustness; class imbalance; per-class recall, hardware latency.
Example video summary: reduces redundancy; risk: gradient descent may miss changes false cuts; verify with temporal consistency.
COMMON MISCONCEPTIONS
HE is not edge detection: it remaps global/local intensities; Sobel/Canny compute spatial gradients.
Chroma subsampling is not compression alone: it reduces samples before/inside coding; codecs also exploit transform, motion, entropy.
Bigger model not always better: deployment constraints and data quality can dominate architecture.
PSNR not enough: high PSNR can still look poor; use SSIM/MOS/task metric.
Open book does not mean copy: answer must apply concepts to scenario; use diagrams/equations compactly.
Do not ignore omitted info: state reasonable assumptions; if impossible, express answer symbolically.
Subsampling notation: 4:J:A describes luma/chroma samples over horizontal groups; exam usually wants sample count and perceptual reason.
Convolution sign: if question says apply masks as written, do not flip; sign of Sobel response depends on convention.

Topic	Can you answer?
storage	derive bytes for RGB/YUV/audio/video
HE	make histogram/CDF/mAP/output
Sobel	compute $G_x, G_y, \nabla $ and interpret
chroma	explain luma+chroma + subsampling
detection	compare VJ/R-CNN/Fast/YOLO + metrics
video	motion estimation + temporal redundancy
compression	JPEG/JPEG2000/codec steps + artefacts
design	pipeline + trade-offs + evaluation

One-line conclusion bank: "This method is suitable because it directly targets [problem] under [constraint], but it may fail when [conditions], so evaluate with [metric]."
Source note: Official 2026 schedule supplies weekly order; local COMPS405 22s1–25s1 finals/recalls supply examinable emphasis and calculation style.
EXAM PHRASEBOOK
Chrominance subsampling: store luma at higher resolution than colour-difference channels because brightness detail dominates perceived sharpness.
Temporal redundancy: adjacent frames often differ by motion + small residual, so store prediction vectors and residual instead of full frames.
Spatial redundancy: neighbouring pixels/transform coeffs correlated; DCT/wavelets compact smooth regions into few low-frequency coeffs.
Coding redundancy: some symbols occur more often; entropy coding gives frequent symbols shorter codes.
Perceptual redundancy: discard/quantise info less visible to HVS or less audible to HAS; risk of artefacts if too aggressive.
CLAHE: local histogram equalisation w/ clip limit; better for uneven illumination than global HE; may amplify local noise.
Canny vs Sobel: Sobel gives gradient estimate; Canny adds smoothing, non-max suppression, hysteresis thresholding for thinner connected edges.
Bilateral filter: weights by spatial closeness + intensity similarity; preserves edges, slower and parameter-sensitive.
Optical flow: per-pixel apparent motion; assumes brightness constancy and small motion; fails under occlusion/lighting change.
Keyframe selection: choose representative frame by uniform sampling, histogram novelty, cluster medoid, or shot boundary; avoid redundant/blurred frames.

SHORT ANSWER SKELETONS
"Explain X" : definition; mechanism; why useful; one limitation; example in digital media.
"Compare A/B" : common goal; key difference; storage/quality/compute effects; when choose each.
"Design solution" : requirements; pipeline; method justification; assumptions; metrics; risks.
"Calculate" : formula; substitution; unit conversion; final; assumption note.
"Evaluate claim" : partly true/false; depends on constraints; counterexample; balanced recommendation.
Fast scoring: use labelled bullets for subparts; show working for every numeric answer; state if using binary MB vs MiB.
Highest priority: raw-size arithmetic; YUV/chroma explanation; HE/Sobel manual calculations; detector pipeline/metrics; integrated design trade-offs.